

COMPUTER VISION - PREDICTIVE MODEL - DATA AVAILABLE

Production Time Estimation by Panel

Casais – Blufab – Modular bathroom panels

CONTEXT

Casais presents this challenge based on the factory operations of BluFab, a Casais Group company focused on the industrialisation and manufacturing of modular bathroom panels and other prefabricated systems. The pilot case is centred on this operational context, but the intended approach should have the potential to evolve and be applied across other industrial companies within the Group.

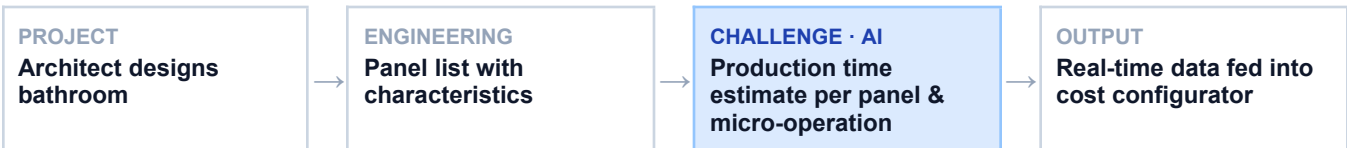
Today, part of the production time estimation logic still depends on manual references, limited measurement exercises, and tacit production knowledge. The challenge is to create a scalable feedback loop between what is designed, what is manufactured, and how long each step actually takes. When estimations fail, the consequences are tangible: commercial proposals misaligned with reality, inaccurate production planning, and difficulty scaling the business based on reliable data.

The expected solution should aim to:

- Detect, in real time or near real time, which task or microtask is being performed at a known workstation using fixed-camera video footage.
- Record the start time, end time, and duration of microtasks within a structured event log aligned with the existing process framework used in BluFab operations (for example: framing, positioning, boarding, screwing, and transitions between stages).
- Associate observed execution with the correct panel and, where feasible, with the corresponding project or production order.
- Use accumulated historical data to predict future manufacturing times before production begins, ideally based on drawings, bills of materials (BOMs), and structured product variables.

In simple terms, Casais does not want video as a passive record. It wants video as operational knowledge: a reusable, searchable, and predictive production layer applied within the BluFab factory environment.

THE FULL FLOW



WHAT TO BUILD

- 1 **Video-based data capture pipeline** — computer vision system that, from a fixed overhead camera, automatically detects transitions between micro-operations and correlates each recording with the corresponding MES production order, without manual intervention. The sequence of operations is predefined and ordered, although not every operation is present for every panel type. The problem is transition detection, not open-ended classification.

- 2 Predictive model of times per micro-operation** — given a panel with its characteristics (cladding type, ceramic, dimensions, drillings), estimate the duration of each micro-operation, based on real production history. The model should be initialised using existing manually compiled Excel estimates and progressively improved as automatically captured production data becomes available.

Computer vision is the data collection mechanism — not the end product. The product is the model that generalises to panels that have not yet been produced, feeding the configurator with real-data-based times.

SCOPE OF THE 48 HOURS

Teams are encouraged to focus on depth rather than breadth during the Fixathon. Recommended scope includes:

- Start with one manufacturing station or one process family, rather than attempting to model the entire factory.
- Work within an existing microtask structure already defined in the BluFab operational context, explicitly identifying at least the main stages considered in the prototype (for example: framing, positioning, boarding, screwing, and transitions between stages).
- Use one or two reference projects as training cases and demonstrate generalisation capability for at least one future panel, variant, or unseen validation case.
- The use of recorded video playback is acceptable during the event, provided the architecture clearly explains the pathway towards real-time or near real-time operation.
- Casais places greater value on reusable data structures and a learning feedback loop than on a visually impressive black-box demonstration.
- 3 test examples will be provided to evaluate the solutions developed

The desired outcome is not merely a prototype, but the foundations of a credible, scalable system capable of transforming operational data into predictive industrial knowledge.

Workstream	Included	Out of Scope / Not Required
Video understanding	Detect microtasks and transitions within a known video context.	Open-ended activity recognition across all workstations and manufacturing contexts.
Time capture	Generate start time / end time / duration records for observed microtasks.	Perfect automation covering all edge cases from day one.
Panel / project linkage	Use available context to assign, prioritise, or flag probable panel/project matches.	Full integration with factory-wide production management.
Prediction	Estimate future production times for at least one unseen or future panel/project.	Production-ready estimation models covering all relevant product families within this operational context.

AVAILABLE DATA FOR FIXATHON

Asset	Content	Why It Matters
Production videos	A small initial set of videos (order of magnitude: from a few videos to a few dozen, to be confirmed before the event), mainly consisting of short recordings of known processes and microtasks.	Primary source for event detection and time capture.

Asset	Content	Why It Matters
MES production orders	List of manufacturing orders with associated panel identification.	Needed for automatic correlation between recording and panel produced.
Microtask / time Excel files	One or more Excel files containing microtask names, process sequences, and manually compiled time estimates.	Reference structure to which solution outputs should be mapped and aligned.
Process drawings	PDFs of manufacturing drawings and process documentation for reference panels and assemblies.	Creates the link between product definition and manufacturing effort.
Bills of Materials (BOMs)	Excel files or structured exports containing the composition of panels/components for reference projects.	Useful for feature extraction and future production time estimation.
Product variables	Product trees, parameters, and structured variables from the BluFab context describing relevant panel and bathroom configuration options.	Supports scalable prediction based on structured product descriptors.
Reference explanations	Short human explanations of one or two projects to help teams interpret process logic.	Improves onboarding and reduces ambiguity.

DATA SAMPLE FOR THE BRIEFING CALL

Production videos Example of one panel video from end to end containing the first phase of a panel construction which covers the process from framing to tiling.	MES production orders List of manufacturing orders as it is available today containing the order number that will be linked to the panel in the sample video.
Microtask / time Excel files Micro-operations identified and recorded based on manual annotation of the sample video.	Process drawings Dimensions, studs, positions, single/double cladding specifications, ceramic types and drillings related to the panel watched in the video.
Bills of Materials (BOMs) Example of a list of materials needed for a panel.	Product variables Example of a product configuration

Privacy: the camera must focus exclusively on the panel. If required, teams should document any proposed face-anonymisation approach.. (it is not an outcome of the fixathon)

EXPECTED OUTPUT PER PANEL

Teams are expected to deliver a functional prototype or proof of concept using real Casais challenge data within the BluFab operational context, producing structured outputs mapped to the process and microtask framework used by Casais. Solutions should demonstrate predictive capability by generating time estimations for at least one future or previously unseen panel/project, based on drawings, bills of materials (BOMs), textual information, and/or other structured variables. Deliverables should also include a simple interface, notebook, dashboard, or API enabling Casais to compare observed versus predicted production times, while understanding the model's confidence levels, assumptions, and limitations.

Field	Source	Description	Format / Notes
MES Order	MES production system	Manufacturing order number linked to the panel being produced	Alphanumeric string. Confirm field name and format used in the MES export.

Field	Source	Description	Format / Notes
Panel ID	Technical drawing	Unique identifier for the panel type, as referenced in the technical drawing	The exact ID field used in the drawings and whether it matches any field in the MES export.
Micro-operation	Micro-operations list	Name of each micro-operation performed on the panel, in sequence order, from the predefined operations list	Only operations applicable to this panel type are listed; absent operations are considered not performed
Duration per micro-operation	CV pipeline / predictive model	Duration in seconds obtained either from CV-based transition detection or from the predictive estimation model, for each applicable micro-operation	Numeric (seconds). One row per micro-operation. Total panel time = sum of all applicable micro-operations.
Total panel time	Derived / aggregated	Sum of all micro-operation durations for the panel; the primary production-time input to the cost configurator	Numeric (seconds). Must be queryable by panel characteristics via the estimation interface.

TECHNICAL ASSUMPTIONS AND TRANSPARENCY REQUIREMENTS

- Teams must clearly distinguish between pre-existing IP/assets, event-specific configurations, and Casais data used within the challenge.
- Teams must briefly document the architecture, model providers, third-party services, licences, hardware/cloud assumptions, and key cost drivers.
- Teams must explicitly indicate what is fully functional, what is heuristic, what is inferred, and what is mocked or manually assisted.
- Portable and replicable approaches are preferred over solutions that depend on privileged or non-reproducible infrastructure.
- Outputs must be exportable (CSV, JSON, API, or equivalent) to support future integration with Casais' systems and decision-making tools, as well as the BluFab ecosystem, with potential for expansion to other industrial operations within the Group.
- Solutions must respect confidentiality obligations and must not expose Casais data in public demos, publications, or reuse beyond the scope of the event.

Casais places greater value on solutions with transparent logic, understandable data structures, and a clear integration pathway than on visually impressive black-box systems that cannot be understood or reused after the event.

PITCH REQUIREMENTS

5 min presentation · 5 min Q&A

Section	What to present	What is evaluated
1 Challenge	Challenge title and the core problem in one sentence. <i>Shared context — no explanation needed. Serves to anchor the pitch.</i>	—

2 Solution and demo	What was built and what it does. Live or recorded demonstration — even if partial. How the user interacts with the solution. <i>This is the centre of the pitch. Showing is more effective than describing.</i>	Direct response to the problem. Quality and completeness of what was delivered. Clarity of the value proposition.
3 Results and evidence	What was tested and with what data. Metrics or observations — even if qualitative. What worked and what did not. Short tech architecture overview and verifiable execution trace <i>Partial results count. The jury wants to see that something was validated, not just built.</i>	Rigour in validation. Honesty about what was and was not tested. Ability to measure impact.
4 Operational viability	Critical dependencies and associated risks. Indicative operating cost (order of magnitude). Main limitations of the current solution. <i>Transparency about limitations is a positive criterion — it does not penalise. Costs as order of magnitude, not detailed budgeting.</i>	Awareness of risks prior to adoption. Realism of cost relative to value generated. Technical maturity of the team.
5 Path to pilot	3–5 concrete steps to move from prototype to real pilot. What needs to be validated first. Minimum resources and who, on the company side, would need to be involved. <i>This is not a product roadmap — it is the immediate, achievable next step.</i>	Implementation-oriented thinking. Understanding of the company's operational context. Credibility and realism of the proposed path.

WHAT WILL BE EVALUATED

Criterion	Weight	What is Success?
Eligibility Threshold	Go/ No-Go	To meet the minimum eligibility threshold, teams must deliver a functional demo using real Casais challenge data, including at least one processed video with a timeline and identified microtask durations, and outputs mapped to the relevant Casais process structure. Solutions should demonstrate initial predictive capability by generating at least one time estimation for a future panel/project or a validation case not used during training. Teams must fully complying with the event's confidentiality and data handling requirements. Solutions based primarily on visual mock-ups, without a functional core pipeline, will not be considered strong submissions.
Predictive model quality	35%	Accurately estimate production time per micro-operation and total panel production time using panel characteristics and historical production data. The solution should demonstrate the ability to generalise to unseen panel configurations and progressively improve as additional production data becomes available.
Technical architecture	20%	Deliver a solution that doesn't trap the challenge owner in any single vendor or platform. Critical dependencies (agents, tools, data stores, models) should be named, isolated behind clear boundaries, and individually swappable. Proprietary primitives are fine when they're the right tool, but any dependency on a vendor's bespoke framework that only runs on their platform must be flagged and justified. The team should also articulate the top architectural risks of their design: what can

Criterion	Weight	What is Success?
		break or fail silently, the main system assumptions, and which of those they chose to accept versus mitigate in the 48 hours available.
Ease / cost of implementation	25%	Be deployable using realistic factory infrastructure constraints, with reasonable hardware, operational and maintenance costs. Preference will be given to solutions requiring minimal manual intervention, limited operational complexity and feasible adoption within BluFab's production environment.
MES integration capability	10%	Reliably correlate production recordings with MES production orders with minimal manual intervention. The integration approach should be practical, resilient to imperfect production data and compatible with existing operational workflows.
Interface usability	10%	Provide a clear and intuitive interface allowing users to query production time estimations, inspect panel-level and micro-operation-level outputs, and understand the model's predictions without requiring technical expertise. Explainability is a core requirement, not a nice-to-have. The solution should communicate not just <i>what</i> the model predicts, but <i>why</i> — for example, which microtasks drive the estimate, how stud layout or panel variant affects predicted time, and where the model is less confident. Solutions that produce accurate estimates but offer no interpretable reasoning will score lower on this criterion.

GLOSSARY

Term	Definition
Cost configurator	The commercial tool used by BluFab to price modular bathroom projects. It receives production time estimates per panel as input. Replacing manually estimated times with real-data-based estimates is the business goal of this challenge.
Frame and clad station	The fixed-table workstation where panel assembly takes place and where the overhead camera is installed. All micro-operations in scope occur at this station.
MES (Manufacturing Execution System)	The factory production management system used by BluFab / Casais. It generates production orders that link each manufacturing run to the panel being produced. The challenge requires automatic correlation between video recordings and MES orders.
Micro-operation / Micro-task	An individual production step at the “frame and clad” station. The full sequence of micro-operations is predefined and ordered; not all operations apply to every panel type. Used interchangeably in this document.
Panel	A modular bathroom component manufactured by BluFab. A single bathroom project is decomposed into up to five panel types, each defined by a technical drawing and a set of characteristics (dimensions, cladding, ceramic, drillings).
Predictive model	The ML model that estimates production time per micro-operation given a panel's characteristics (dimensions, cladding, ceramic, drillings). It is initialised with manually compiled Excel data and progressively improved as CV-captured data accumulates.

Term	Definition
Stud Layout	The arrangement and spacing of vertical structural members (studs) within a modular construction panel. In this challenge, stud layout determines the sequence and complexity of assembly microtasks, and is a key variable in predicting panel assembly time.
Technical drawing	Engineering document that fully specifies a panel: dimensions, stud layout, cladding type (single/double), ceramic specification, and drilling positions. Provided for all panels in scope; data is considered accurate.
Transition detection	The CV sub-problem of identifying the moment when one micro-operation ends and the next begins. Because the sequence is fixed and known, the system does not need to classify which activity is occurring — only when it changes.

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